Buzz: Build and Change Report

AI Reel Studio: reels from photos, videos or just an idea | version 2.0.0 | 12 Nov 2026

Buzz turns a stack of photos and videos, or just a written idea, into a ready-to-post 9:16 Instagram reel with music, text, a caption and five hashtags. It runs on a machine you control and works on its own in the background.

At a glance

Size	30 Python files, about 5,084 lines
Runs on	Windows 10 or 11, CPU only
Engine	Python 3.12, FFmpeg, librosa, OpenCV, Streamlit; Claude for understanding and planning
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What it does

Runs anywhere	Windows PC, Linux PC or a Linux server, with the same features. Several people can use one server.
Your reels, your key	Everyone sees only their own reels and can use their own Claude key. The owner sets limits on the shared key and has an emergency pause switch.
Two ways to start	Upload photos and videos (with your own song, music made for you, or silent), or type an idea and get a finished reel from the words alone.
Smart editing	The AI studies your pictures and clips, plans a story, cuts on the beat, keeps text clear of faces, and lets you choose the opening and closing shot.
Instagram pack	Hook, caption, 5 hashtags, cover text, best time to post and alt text for every reel.
Idea reels	Ten colour themes, with soft gradient, plain colour, drawn pictures or stock video behind the words.
Voiceover	A free voice that runs on your PC (Kokoro), or Google Cloud Text-to-Speech with your own key. Music dips under the voice.
Your control	Reel length from 15 to 90 seconds, how many pictures to use (Auto, All or a number), change notes in plain words, optional logo.
Looks after itself	Live progress, a background worker that restarts itself, accounts with sign-in and remember-me, safe uploads, and a nightly clean-up of old reels.

What you need

- Windows 10 or 11 with a normal user account (no administrator needed), about 4 GB of free disk, and internet access.
- A Claude API key from console.anthropic.com (required). Buzz is powered by Claude.
- Optional: a Google Cloud Text-to-Speech key (voiceover with Indian, UK and US voices), a Pexels key (stock video for idea reels), a mailbox for emailing password-reset codes.

Change history

2.0.0 | 12 Nov 2026 | Linux, servers and several users

- Linux version: installer (install_linux.sh), uninstaller, systemd services with CPU and memory caps, nginx file with websocket support, server guide (docs/SERVER.md). Tested on Ubuntu 24.04 including a from-scratch install.

- Cross-platform code: encrypted secret store on Linux (Windows Credential Manager on Windows), picture drawing without a browser (resvg), free fonts bundled for Linux, worker control without Windows Task Scheduler.
- Per-user reels: members see only their own reels; the owner sees all.
- Per-user Claude keys, usage tracking (tokens only), per-person monthly token and daily reel limits, a studio monthly budget, optional cost estimates and a Pause AI for everyone switch.
- The free voice now runs in its own helper process, so its memory (about 1 GB) returns to the system after every reel; it restarts by itself if it ever stops.
- Line endings of scripts are fixed to Unix; automated tests for the per-user rules (tests/test_per_user.py).
- Server lifecycle verified on Ubuntu 24.04: services under a dedicated user restart within seconds after a crash and the interrupted reel resumes; the nginx proxy passes websockets; the uninstaller removes services and environment but keeps reels, accounts and keys.

1.0.9 | 12 Nov 2026 | README accuracy

- README: the Smart App Control section now states only what Microsoft documents (link added) and no longer mentions evaluation mode or planned signing.

1.0.8 | 12 Nov 2026 | Uninstaller

- New Uninstall Buzz.bat: removes the background tasks and the Python environment, and separately asks before removing saved keys or deleting the whole folder (reels and accounts are kept by default).
- README: clearer guidance for Windows 11 Smart App Control, without asking anyone to lower a security setting.

1.0.7 | 12 Nov 2026 | Key entry

- Saving the Claude key now checks it with Claude straight away: a wrong key is refused with a clear message and nothing is saved.
- After a key is saved the box clears and Save is greyed out until you type a new key, so it is obvious the key was stored.

1.0.6 | 12 Nov 2026 | Lighter Claude connection

- Buzz now talks to Claude directly through a small plain-Python client (claude_api.py) instead of the Anthropic SDK. This removes the SDK and its compiled helper (jiter), which Windows Smart App Control can block, so reels can be made on those PCs.
- Clearer messages when Claude refuses a key or cannot be reached; automatic retries when Claude is busy.

1.0.5 | 12 Nov 2026 | Installer window

- The installer window now closes by itself after a successful install, and waits for a key press only if something went wrong.

1.0.4 | 12 Nov 2026 | Smoother flow

- After you press Create or Upload, the page moves to My reels by itself and shows the new reel being prepared.
- README: a note for Windows 11 Smart App Control, which can block unsigned Python library files.

1.0.3 | 12 Nov 2026 | Installer fix

- Fixed the background-task step of the installer failing on some Windows setups (a PowerShell variable named args clashed with a built-in).
- The installer now keeps a log (Work\install.log) and, if anything fails, shows a clear message and waits instead of closing.

1.0.2 | 12 Nov 2026 | Key entry moved into the app

- The installer no longer asks for the Claude key. You add it on the Settings tab, and a banner reminds you until it is saved.
- README: how to get past Windows 11 Smart App Control (unblock the zip before extracting).

1.0.1 | 12 Nov 2026 | Install location

- The installer asks where to install (default: Documents\Buzz), copies Buzz there and sets everything up in that folder. No fixed paths are built in.
- Updating: run the new installer and choose the same folder; reels, accounts and settings are kept.

1.0.0 | 12 Nov 2026 | First public release

- Two modes in one app: reels from photos and videos, and reels from only an idea.
- AI understanding of every picture and clip, beat-matched cutting, face-safe text, collages for landscape pictures, optional portrait centring.
- Choose the opening and closing picture or clip; the story is built around them.
- Music: use your own song, or generate original music from up to three keywords, tuned to be clear on phone speakers.
- Voiceover: free local voice (Kokoro) or Google Cloud Text-to-Speech; background music ducks under the voice.
- Idea reels: ten colour themes; gradient, plain colour, drawn pictures or Pexels video backgrounds.
- Instagram pack: hook, caption, 5 hashtags, cover text, best time, alt text; download as text; readable file names.
- Reel length selector (15 to 90 s) and an assets selector (Auto, All, or a number).
- Change notes in plain words ('make shot 5 the opening', 'use a different font') and one-tap quick changes.
- Accounts with email and password, 30-day remember-me, forgot-password codes, owner and member roles.
- Settings: Claude key, free voice install, Google voice key, Pexels key, mail, system status, automatic cleanup.
- Safe uploads: file-type and content checks, size limits, no executable files.
- Live page: auto-refreshing lists and a smooth real-time progress bar.
- Automatic cleanup: reels older than 15 days move to a holding folder and are deleted 3 days later; Keep switch per reel.
- My reels shows the last 15 days.
- Video encoding uses an NVIDIA, Intel or AMD graphics card automatically when one is present and its quality at least matches the CPU encoder; otherwise the CPU.
- Windows installer (no administrator needed) that sets up Python, FFmpeg, the environment and two background tasks.

Modules

worker.py / service_ctl.py	Background worker: one job at a time, restarts itself; start and stop on any system
secrets_store.py	Where keys are kept: Credential Manager on Windows, an encrypted file on Linux
usage.py	Reel owners, per-user keys, usage tracking and limits
pipeline.py	One folder of media to one finished reel
ingest.py	Reads and catalogues photos and videos
music.py / music_provider.py	Beat and energy analysis; original music generator
ai_plan.py	AI understanding, story plan, asset count, opening and closing shot
render.py / encoder.py	FFmpeg rendering, text, logo, audio mix; picks the CPU or a graphics-card video encoder by testing quality
layout.py / facedetect.py	Keeps text clear of faces
revise.py	Plain-language change notes
prompt_reel.py / art_provider.py	Idea reels: script, scenes, themes, drawn pictures
claude_api.py	Plain-Python Claude Messages client with retries
voice_provider.py / voice_worker.py / narration.py	Kokoro (in its own helper process) and Google voices, voiceover timing

stock_provider.py	Pexels stock video
social.py	Instagram pack
progress.py	Real-time progress
uploads.py	Safe upload checks
auth.py / mailer.py	Accounts, sessions, reset codes, optional email
cleanup.py	Nightly tidy-up of old reels
app.py	The web page

Keys used

anthropic	Claude key (required)
google_tts	Google Text-to-Speech key (optional)
pexels	Pexels key (optional)
smtp	Mail settings for reset codes (optional)

Buzz is powered by Claude. The code is open: you are free to change it and plug in the AI of your choice, whichever works for you.